TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	76.8 ± 2.27
Correlation Stability	88.1 ± 1.76
Distribution Consistency	84.4 ± 1.85
Entropy Preservation	86.0 ± 1.10
Mean Z Score Drift	0.2 ± 0.02
Categorical Fidelity	82.1 ± 0.54
Privacy Risk	50.4 ± 1.28
Data Density	99.9 ± 0.10
Coverage Score	19.3 ± 4.96
Feature Accuracy	77.9 ± 2.77
Novelty Score	0.5 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	40.629	-4.117	9.2%	OK
TB	3.299	1.778	-1.521	46.1%	Drift
DB	1.486	0.738	-0.748	50.3%	Drift
Alkphos	290.576	245.536	-45.040	15.5%	Warn
Sgpt	80.714	49.491	-31.222	38.7%	Drift
Sgot	109.911	55.590	-54.321	49.4%	Drift
TP	6.483	6.868	+0.385	5.9%	OK
ALB	3.142	3.500	+0.358	11.4%	Warn
A/G Ratio	0.947	1.088	+0.141	14.8%	Warn
Selector	1.286	1.369	+0.083	6.5%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	76.8 ± 2.27
Correlation Stability (*2)	88.1 ± 1.76
Distribution Consistency (*3)	84.4 ± 1.85
Entropy Preservation (*4)	86.0 ± 1.10
Mean Z Score Drift (*5)	0.2 ± 0.02
Categorical Fidelity (*6)	82.1 ± 0.54
Privacy Risk (*7)	50.4 ± 1.28
Data Density (*8)	99.9 ± 0.10
Coverage Score (*9)	19.3 ± 4.96

Feature Accuracy (*10)	77.9 ± 2.77
Novelty Score (*11)	0.5 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.5683760683760684	0.7102564102564102	0.00427350427350428	
F1	0.8128002697439015	0.6603616007724427	0.8103099231232467	-0.0024903466206549043	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6931623931623931	0.5350427350427351	0.6606837606837607	-0.03247863247863249	
F1	0.79586898930459	0.6414824179033258	0.7620880021155565	-0.033780987189033454	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6982905982905983	0.7068376068376068	0	
F1	0.8279091624529537	0.8215757501493922	0.8279091624529537	0	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.647008547008547	0.6811965811965812	-0.00512820512820511	
F1	0.7971299319768493	0.7595758773049185	0.7950609371186629	-0.002068994858186479	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.5547008547008547	0.6931623931623931	-0.009401709401709401	
F1	0.7997903571549948	0.6389199960995223	0.7901731751006158	-0.009617182054378887	

Model: NB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.5256410256410257	0.5598290598290598	0.00085470085470085	
F1	0.5587432119380917	0.6408116237470537	0.5602601611890189	0.0015169492509272964	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.5675213675213676	0.6965811965811965	-0.010256410256410243	
F1	0.8097855895248811	0.657429266568766	0.8014242018782506	-0.008361387646630502	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6589743589743591	0.4726495726495764	0.6555555555555557	-0.003418803418803488	
F1	0.76253448323086	0.5620733964138589	0.7544345379927289	-0.008099945238131146	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6829059829059829	0.5230769230769231	0.6811965811965812	-0.0017094017094017033	
F1	0.7891849307093717	0.6026999283696599	0.7848020165423577	-0.004382914167013929 5	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.5435897435897437	0.6829059829059829	0.00854700854700854	
F1	0.7821343746683586	0.6464939042205462	0.7946306638041012	0.012496289135742545	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	44	Male	0.8	0.2	335	148	86	5.6	3.0	1.1
2	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06
3	50	Male	0.7	0.2	188	12	14	7.0	3.4	0.9
4	18	Male	1.4	0.6	215	440	850	5.0	1.9	0.6
5	60	Male	0.7	0.2	171	31	26	7.0	3.5	1.0
6	25	Male	0.8	0.1	130	23	42	8.0	4.0	1.0
7	29	Female	0.9	0.3	150	35	40	6.5	3.2	1.0
8	52	Male	1.5	0.6	220	60	70	7.0	3.8	2.0
9	45	Female	0.8	0.2	180	25	30	5.8	2.9	1.0
10	38	Male	2.1	0.9	250	45	50	6.9	3.5	1.0
11	60	Female	1.2	0.4	300	50	55	7.5	4.0	2.0
12	34	Male	0.7	0.2	190	20	25	6.0	3.0	1.0
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Male	1.2	0.4	282	36	32	7.2	3.9	1.1
2	66	Male	0.7	0.2	239	27	26	6.3	3.7	1.4
3	27	Male	1.2	0.4	179	63	39	6.1	3.3	1.1
4	31	Female	0.8	0.2	158	21	16	6.0	3.0	1.0
5	6	Male	0.6	0.1	289	38	30	4.8	2.0	0.7
6	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
7	34	Female	1.0	0.3	210	22	30	6.5	3.2	0.8
8	47	Male	2.1	0.5	180	35	40	7.0	3.8	1.2
9	29	Female	0.9	0.2	150	18	25	6.0	3.1	0.9
10	52	Male	3.5	1.2	250	45	55	7.4	4.0	1.0
11	63	Female	0.8	0.1	190	20	28	6.8	3.5	1.1
12	41	Male	1.4	0.6	220	30	35	7.2	3.9	1.3
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	52	Male	0.6	0.1	178	26	27	6.5	3.6	1.2
2	4	Male	0.8	0.2	460	152	231	6.5	3.2	0.9
3	43	Female	0.9	0.3	140	12	29	7.4	3.5	1.8
4	60	Male	8.9	4.0	950	33	32	6.8	3.1	0.8
5	50	Male	0.9	0.3	901	23	17	6.2	3.5	1.2
6	36	Male	1.7	0.5	205	36	34	7.1	3.9	1.2
7	29	Female	1.2	0.4	210	45	30	7.5	3.8	1.1
8	37	Male	0.9	0.3	180	25	22	6.8	3.4	0.9
9	54	Female	2.5	1.0	300	50	40	7.0	3.2	0.8
10	41	Male	3.0	1.5	250	60	50	8.1	4.0	1.2
11	48	Female	0.7	0.2	190	20	15	6.5	3.1	0.9
12	34	Male	1.8	0.7	220	35	28	7.2	3.6	1.0
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	46	Male	0.7	0.2	224	40	23	7.1	3.0	0.7
2	18	Male	1.8	0.7	178	35	36	6.8	3.6	1.1
3	42	Female	7.4	3.6	298	52	102	4.6	1.9	0.7
4	22	Male	2.7	1.0	160	82	127	5.5	3.1	1.2
5	36	Female	0.8	0.2	650	70	138	6.6	3.1	0.8
6	37	Male	0.8	0.2	147	27	46	5.0	2.5	1.0
7	29	Female	1.2	0.5	210	45	50	6.5	3.2	0.9
8	34	Male	0.9	0.3	175	60	70	7.0	3.5	1.0
9	41	Female	2.5	1.1	300	55	80	6.8	3.4	0.8
10	27	Male	1.0	0.4	190	50	60	7.2	3.1	0.7
11	38	Female	0.8	0.2	220	40	45	5.9	2.8	0.6
12	55	Male	3.2	1.5	400	70	90	7.5	3.6	1.1
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	20	Male	1.1	0.5	128	20	30	3.9	1.9	0.95
2	62	Male	0.7	0.2	173	46	47	7.3	4.1	1.2
3	48	Female	1.0	0.3	310	37	56	5.9	2.5	0.7
4	55	Male	0.8	0.2	290	139	87	7.0	3.0	0.7
5	75	Male	0.9	0.2	282	25	23	4.4	2.2	1.0
6	37	Male	0.8	0.2	195	60	40	8.2	5.0	1.5
7	29	Female	0.8	0.3	150	25	30	6.5	3.2	1.1
8	34	Male	1.2	0.5	200	40	50	7.0	3.8	0.9
9	45	Female	0.6	0.2	180	20	25	5.5	2.8	0.8
10	52	Male	1.5	0.7	220	60	70	8.0	4.0	1.0
11	39	Female	0.9	0.3	160	30	35	6.0	3.0	1.2
12	27	Male	1.1	0.4	190	50	45	7.5	3.5	0.95
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	31	Male	0.6	0.1	175	48	34	6.0	3.7	1.6
2	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
3	73	Male	1.8	0.9	220	20	43	6.5	3.0	0.8
4	38	Female	0.8	0.2	185	25	21	7.0	3.0	0.7
5	45	Male	0.7	0.2	180	18	58	6.7	3.7	1.2
6	55	Male	0.8	0.2	482	112	99	5.7	2.6	0.8
7	29	Female	1.2	0.5	150	30	25	6.5	3.8	1.7
8	62	Male	2.5	1.0	300	45	40	7.0	4.0	1.2
9	34	Female	0.9	0.3	175	22	30	6.8	3.5	1.0
10	48	Male	1.8	0.9	250	50	55	7.2	3.9	1.3
11	55	Female	0.7	0.2	160	20	28	6.0	3.2	0.9
12	40	Male	3.0	1.5	400	60	70	8.0	4.5	1.0
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	6	Male	0.6	0.1	289	38	30	4.8	2.0	0.7
2	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
3	32	Male	23.0	11.3	300	482	275	7.1	3.5	0.9
4	21	Female	0.6	0.1	186	25	22	6.8	3.4	1.0
5	8	Female	0.9	0.2	401	25	58	7.5	3.4	0.8
6	60	Male	2.4	1.0	1124	30	54	5.2	1.9	0.5
7	25	Female	1.2	0.4	150	22	18	6.5	3.2	0.8
8	34	Male	2.5	1.0	200	45	30	7.0	3.5	0.9
9	42	Female	0.8	0.3	120	30	25	6.0	2.9	0.7
10	29	Male	3.0	1.2	250	50	40	7.5	3.8	1.0
11	55	Female	1.5	0.5	180	35	20	6.8	3.4	0.9
12	48	Male	2.2	0.9	300	60	45	8.0	4.0	1.1
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	29	Female	0.7	0.1	162	52	41	5.2	2.5	0.9
2	23	Male	1.0	0.3	212	41	80	6.2	3.1	1.0
3	51	Male	2.9	1.3	482	22	34	7.0	2.4	0.5
4	36	Female	0.7	0.2	152	21	25	5.9	3.1	1.1
5	70	Male	1.3	0.4	358	19	14	6.1	2.8	0.8
6	55	Male	1.1	0.3	215	21	15	6.2	2.9	0.8
7	34	Female	0.8	0.3	210	45	30	6.5	3.1	1.0
8	29	Male	1.2	0.4	180	50	40	7.0	3.5	0.9
9	62	Female	2.5	1.1	300	60	50	7.8	4.0	1.2
10	40	Male	1.0	0.2	250	30	20	6.2	2.8	0.8
11	55	Female	0.9	0.3	190	35	25	6.0	3.2	0.95
12	48	Male	3.0	1.5	400	70	60	8.0	3.8	1.1
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	45	Male	1.1	0.4	92	91	188	7.2	3.8	1.11
2	36	Female	0.7	0.2	152	21	25	5.9	3.1	1.1
3	70	Female	0.9	0.3	220	53	95	6.1	2.8	0.68
4	50	Male	1.2	0.4	282	36	32	7.2	3.9	1.1
5	33	Male	0.8	0.2	135	30	29	7.2	4.4	1.5
6	50	Male	1.6	0.8	218	18	20	5.9	2.9	0.96
7	34	Female	1.2	0.5	145	30	40	6.8	3.5	0.85
8	29	Male	0.9	0.3	210	25	35	7.0	3.2	0.75
9	52	Female	2.5	1.0	300	50	45	7.5	4.0	0.95
10	41	Male	1.0	0.4	180	20	30	6.5	2.9	0.78
11	60	Female	0.8	0.2	160	15	20	5.9	3.1	0.89
12	38	Male	3.0	1.5	220	100	150	8.0	4.5	1.1
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	0.8	0.2	158	27	23	6.7	3.1	0.8
2	54	Male	0.8	0.2	218	20	19	6.3	2.5	0.6
3	60	Male	1.8	0.5	201	45	25	3.9	1.7	0.7
4	73	Male	1.9	0.7	1750	102	141	5.5	2.0	0.5
5	46	Male	0.8	0.2	160	31	40	7.3	3.8	1.1
6	60	Male	0.7	0.2	174	32	14	7.8	4.2	1.1
7	34	Female	0.9	0.3	150	25	30	6.5	3.2	1.1
8	45	Male	1.5	0.5	200	40	50	7.0	3.5	1.2
9	29	Female	0.8	0.2	180	20	25	6.0	3.0	1.0
10	52	Male	2.3	1.0	220	55	60	7.5	3.8	1.3
11	38	Female	1.1	0.4	160	30	35	6.8	3.4	1.2
12	41	Male	0.7	0.2	190	32	40	6.3	3.1	1.0
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3